National College of Ireland

Project Submission Sheet

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Student ID:	24235415		
Programme:	Msc in Cloud Computing	**Year:**	2025-2026
Module:	Fog and Edge Computing		
Lecturer:	Dr. Diego Lugones		
Submission Due Date:	17st April 2026		
Project Title:	Smart Chair Health Monitoring System: Fog and Cloud IoT Architecture on AWS		
Word Count:	……		

Academic Integrity Certificate Verification Code: Q6lveHvE6b

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Signature:	Meenakshi ……
Date:	17st April 2026 ……

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3. **You must ensure that you retain a HARD COPY of ALL projects**, both for your own reference and in case a project is lost or mislaid. It is not sufficient to keep a copy on computer. Please do not bind projects or place in covers unless specifically requested.

4. You must ensure that all projects are submitted to your Programme Coordinator on or before the required submission date. **Late submissions will incur penalties.**
5. All projects must be submitted and passed in order to successfully complete the year. **Any project/assignment not submitted will be marked as a fail.**
6. **Please check that you read AI and Academic Integrity Acknowledgement Supplments in this document**

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[Insert Module Name]

[Insert Title of your assignment]

Your Name/Student Number	Course	Date
Meenakshi	Msc in Cloud Computing	**17st** April 2026

This section is a supplement to the main assignment, to be used if AI was used in any capacity in the creation of your assignment; if you have queries about how to do this, please contact your lecturer. For an example of how to fill these sections out, please click here.

AI Acknowledgment

This section acknowledges the AI tools that were utilized in the process of completing this assignment.

Tool Name	Brief Description	Link to tool

Description of AI Usage

This section provides a more detailed description of how the AI tools were used in the assignment. It includes information about the prompts given to the AI tool, the responses received, and how these responses were utilized or modified in the assignment. **One table should be used for each tool used**.

[Insert Tool Name]	
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The red square below indicates certificate number, please put this in the above field



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Smart Chair Health Monitoring System: Fog and Cloud IoT Architecture on AWS

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***Abstract*—This Project Focuses On Scable fog to cloud architecture of IoT to monitor human health condition in office environment. The Smart Chair Health Monitoring System gathers sensor information from a simulated chair that measures seat pressure, heart rate, posture angle, body temperature, and movement. Data gathered from these sensors is processed at the Fog node that runs on an EC2 instance and then sent to the cloud backend. The backend architecture consists of using the AWS API Gateway, Lambda function, and DynamoDB as storage services to process the sensor data. Also, the system includes a real-time dashboard that displays sensor readings through the use of HTML, JavaScript, and Chart.js and triggers notifications in case of anomalies such as poor posture or lack of physical activity. This project shows how fog computing and cloud computing are applied for a scable IoT system. In this way, latency is minimized in addition to automatic scalability for real-time application. Keywords: Fog Computing, IoT, AWS, DynamoDB, Serverless Architecture, Health Monitoring**

I. Introduction

Employees in offices have to spend So much of hours sitting at their desks working on computers, which results in problems related to posture, spine, and physical exercise. Ergonomic monitoring systems can be used for detection of unhealthy behavior and alerting about the need to take actions. In terms of IoT solutions, sensors can be used to detect the current physical situation. At the same time, transmitting all collected data to the cloud can cause delay in data analysis and higher load on network resources. To solve this problem, fog computing is used. The goal of the Smart Chair Health Monitoring System project is the development of monitoring system based on fog-to-cloud approach. It will be developed for detecting and storing health data of office environment using virtual sensors that create ergonomic data. The following tasks need to be performed within this project: 1. Collect data from several simulated sensors. 2. Create fog node. 3. Create scalable backend based on AWS cloud services. 4. Store processed data in DynamoDB database. 5. Visualize sensor data in a dashboard.

II. Architecture and Design

The system follows a three-layer architecture consisting of sensors, fog nodes, and cloud services.

Component	Role	Technology
Sensors	Generates chair health data	Python simulator
Fog Node	Data processing and alert detection	AWS EC2 + Flask
API Gateway	Public cloud interface	AWS API Gateway
Data Processing	Event processing	AWS Lambda
Storage	Sensor data storage	DynamoDB
Dashboard	Data visualization	HTML + Chart.js
Terraform	Automates the process of resource creation	IaC tool

Fig. 1. Component and its role

System Data Flow Diagram 1. Sensor simulator produces health monitoring data.
2. This data is transmitted to the fog node through HTTP calls.
3. The fog node analyzes the data and sends alerts.
4. Analyzed data is then passed to the API Gateway.
5. Lambda get invoked and stores the data in DynamoDB.
6. The dashboard gets the data from the backend system and visualizes it and gives us the graph and live data for the chair which we are tracking.

III. Application Description

The Smart Chair system monitors the human body conditions using five simulated sensors.

Sensor types used

Key Features

Multi-Sensor Monitoring

The system collects data from multiple sensor types simultaneously.

Real-Time Alerts

Alerts are generated when abnormal conditions are detected and alerts are generated.

Example: posture angle ¿ 40° - Bad posture alert movement = false - inactivity alert

Real-Time Dashboard

The dashboard displays live sensor readings and graphical trends using Chart.js.

Design Justification Terraform

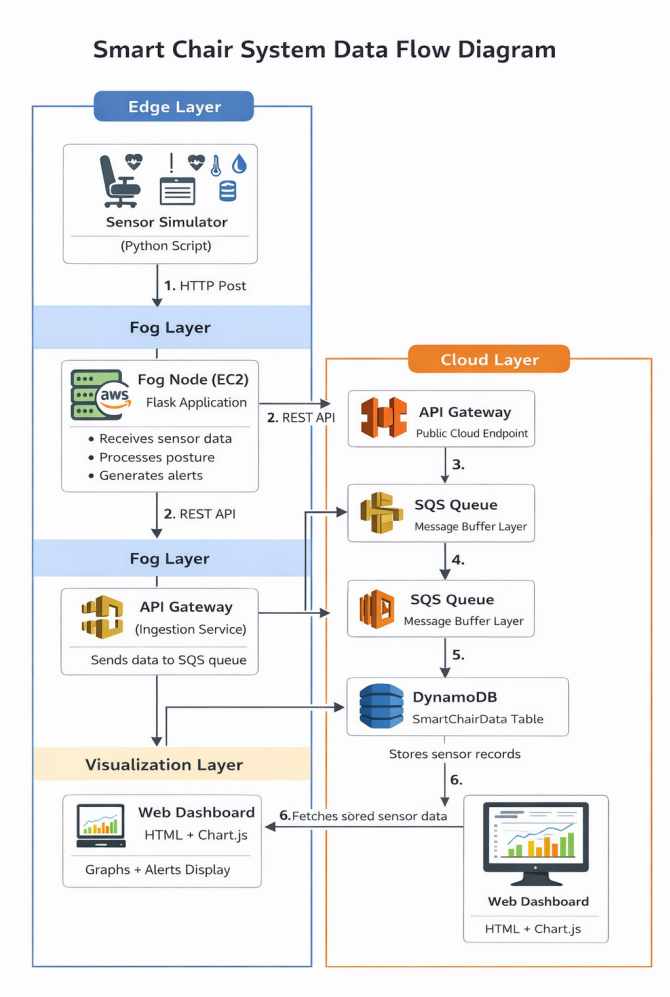


Fig. 2. Architecture Diagram

Sensor	Description
Seat Pressure	Detects sitting pressure levels
Posture Angle	Measures back posture
Heart Rate	Monitors user heart rate
Temperature	Detects body temperature
Movement	Detects inactivity

Fig. 3. Table of Sensor Used

Its and IAC tool for provisioning the resources which has been used during the testing and overall experiment.

Fog Layer

The fog node is installed on an EC2 instance and serves as a middleware layer for processing sensor data before transferring it to the cloud backend. This architecture lowers the amount of time required for data transfer as all processing takes place

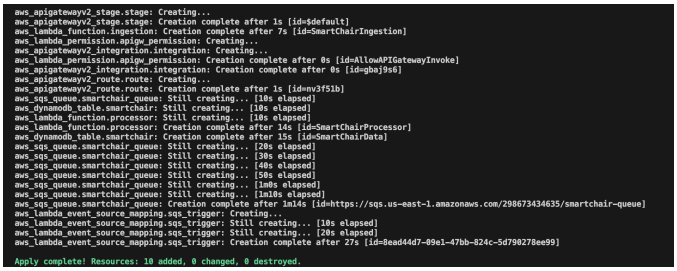


Fig. 4. terraform implementation

locally before being transferred to the cloud backend.

Serverless Processing

AWS Lambda is used to process incoming events from API Gateway. Lambda automatically scales based on incoming traffic and eliminates the need for manual server management.

Cloud Storage

DynamoDB is chosen as a backend database because of its ability to scale rapidly along with time-series data and perform quick queries. Dashboard

Dashboard Visualization

Chart.js is utilized to provide visual representations of sensor data using JavaScript.

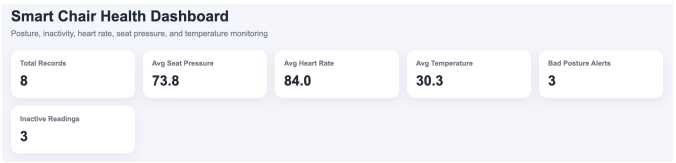


Fig. 5. Sensor Data

IV. Implementation

The system implementation consists of three main components.

Sensor Simulation

The sensor simulator generates random values representing Human body posture conditions. Example payload:

”chair_id”: ”chair-01”,
”seat_pressure”: 75,
”posture_angle”: 50,
”heart_rate”: 85,
”temperature”: 32,
”movement”: false

Processing at the Fog Node The processing at the fog node involves running a Flask application on the EC2 instance. The sensor data is collected and processed, which involves data validation and alert generation. Example for alert generation:

if posture angle greater than 40:
alert = ”Bad posture”

Cloud Backend The backend architecture consists of API Gateway (smartchair-api), Lambda (SmartChairProcessor,SmartChairIngestion) and DynamoDB(SmartChairData).

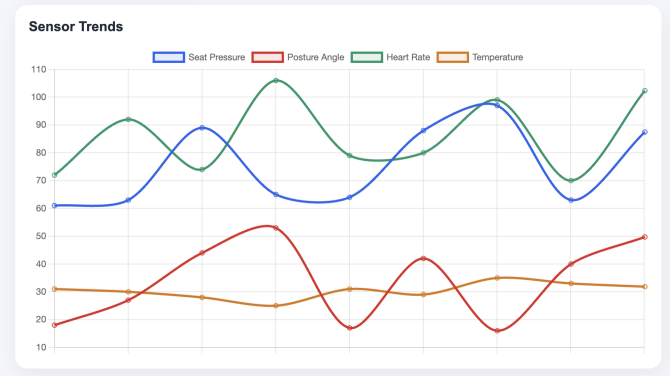


Fig. 6. Sensor Trends

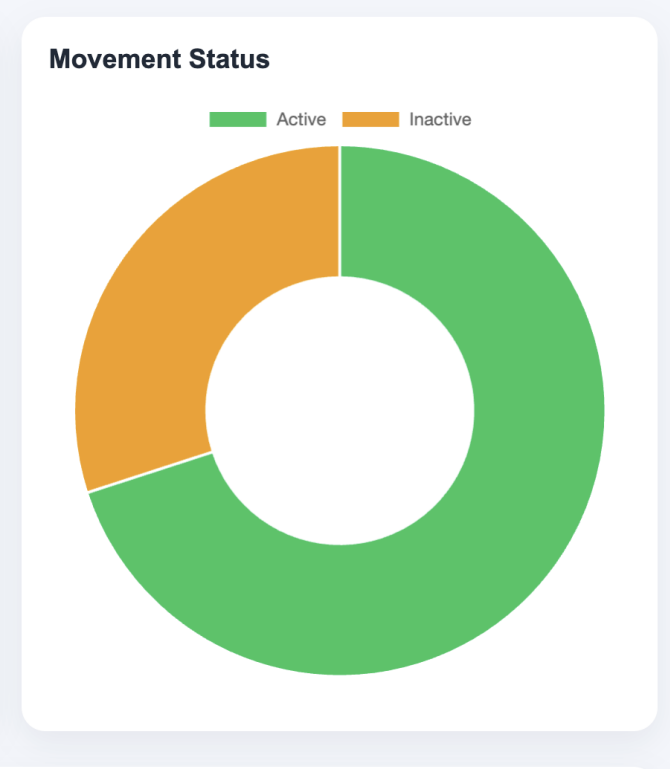


Fig. 7. Movement Status

Lambda functions receive data from API Gateway and store the processed sensor readings in DynamoDB.

Dashboard Implementation

The dashboard takes data from the backend API and displays:

- sensor trends
- alert notifications
- live fetched readings

Graphs are generated using Chart.js.

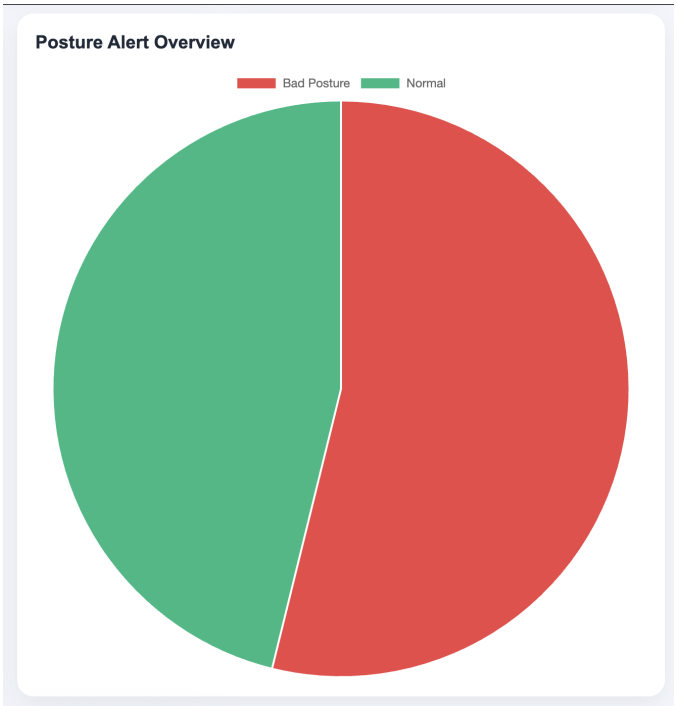


Fig. 8. Posture alert overview

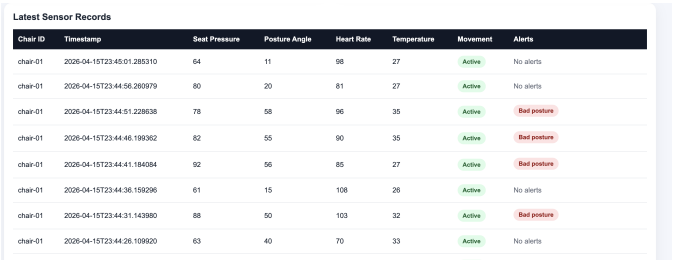


Fig. 9. Latest Sensor Records

V. Results

The system successfully processes real-time sensor data and displays results through the dashboard.

Key outcomes include:

- Real-time data visualization
- Alert detection for poor posture
- Cloud-based data storage
- Scalable architecture

The system demonstrates effective integration between fog computing and cloud services.

VI. Conclusion and Future Work

The above test is a working example of how to implement the use of a scalable IoT approach in monitoring the ergonomic environment in workplaces. In this context, sensor simulation, fog computing, cloudless computation, and real-time dashboard are all achieved. By utilizing fog computing and cloud services, we have created a highly efficient

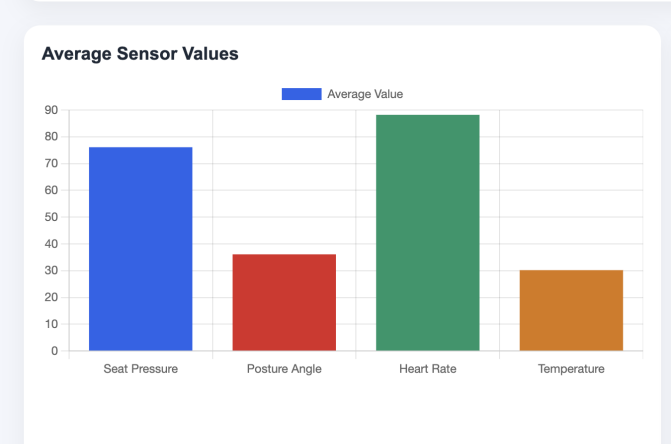


Fig. 10. Average Sensor Data

chair_id (String)	timestamp (String)	alerts	heart_rate	movement	posture_ang
chair-01	2026-04-16T14:45:17....	[{ "S" : "Use...	106	false	37
chair-01	2026-04-16T14:45:23....	[{ "S" : "Use...	95	false	30
chair-01	2026-04-16T14:45:28....	[{ "S" : "Bad...	88	true	55
chair-01	2026-04-16T14:45:33....	[]	75	true	18
chair-01	2026-04-16T14:45:38....	[]	78	true	10
chair-01	2026-04-16T14:45:43....	[]	85	true	39
chair-01	2026-04-16T14:45:49....	[{ "S" : "Bad...	74	true	49
chair-01	2026-04-16T14:45:54....	[{ "S" : "Use...	102	false	40

Fig. 11. Dynamo Data

architecture without any latency issues and maintaining scalability. The following could be some of the future improvements: Implementation of hardware sensors
Notification mechanism for portable devices
Predictive machine learning based on posture
Multichair capability for larger office spaces

[1] [2] [3] [4] [5] [6] [7] [8] [9] [10] [11]

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